

PE-RankFormer: A Minimally-Mechanistic Relational Transformer for Prime-Editing Efficiency Prediction

PEFormer research pilot

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Abstract

Prime editing (PE) efficiency prediction has been dominated by mechanistic models that hard-code assumptions about the PE reaction pathway (nick, flap resolution, mismatch repair, etc.), most recently OptiPrime (Hsu et al., 2026), a strongly mechanistic model trained on 297,962 PE experiments. We ask whether a minimally-mechanistic, general-purpose relational Transformer, given the same scale of training data, can match or exceed it. We reconstruct 285,260 of the 297,962 target training examples (95.7%) from four source studies, implement PE-RankFormer – a 26.6M-parameter paired-sequence Transformer with cross-attention between the edit-outcome encoding and the pegRNA encoding and FiLM-based conditioning on experimental context – and train it with a joint regression and within-target ranking objective. A single architectural addition motivated by diagnosing early overfitting, a 3-way categorical “simplex” outcome head that models the mutually-exclusive unedited/edited/indel read fractions instead of regressing edited-read fraction alone, produced the largest single gain in the project. An ensemble of PE-RankFormer variants, evaluated head-to-head against OptiPrime’s own released 5-model ensemble on the 15,022-row Hsu subset of the locked test fold, achieves Spearman $\rho = 0.794$ versus OptiPrime’s 0.724 (Pearson 0.767 vs. 0.724), a difference of +0.070 with a protospacer-clustered paired bootstrap 95% CI of [+0.036, +0.105] ($p < 0.0005$, PE-RankFormer wins in 2000/2000 resamples). The same comparison run on an earlier, less complete 262,508-row reconstruction gives $\rho = 0.807$ (+0.083, CI [+0.047, +0.118]), so the conclusion is robust to a 22,752-row difference in corpus composition. Notably, PE-RankFormer wins while training on strictly *less* data than OptiPrime in every partition, and while holding out a test fold that OptiPrime had itself trained on. We report these results together with negative results from the same sweep and explicit caveats about corpus completeness and test-fold query discipline.

Contents

1	Introduction	1
2	Training Corpus Reconstruction	2
3	Model Architecture	3
3.1	The simplex outcome head: diagnosis-driven innovation	3
4	Training Procedure	4
5	Ablation and Configuration Sweep	4

6	Headline Result: Ensemble vs. Ensemble Comparison with OptiPrime	6
6.1	Statistical significance	7
6.2	Effect of corpus expansion	7
6.3	Full test fold metrics	7
7	Discussion and Caveats	10
8	Conclusion	11

1 Introduction

Prime editing efficiency prediction is typically approached with strongly mechanistic models: pipelines that explicitly encode the PE reaction as a sequence of biochemical steps (spacer/PBS/RTT annealing, flap competition, mismatch-repair outcome) and use that structure to constrain the model. OptiPrime (Hsu et al., 2026) is the current state of the art on this axis, trained on 297,962 PE experiments assembled across four studies.

The research question motivating this pilot is whether a *minimally mechanistic* model – one that encodes only the raw sequences and experimental metadata and lets a general-purpose relational architecture (a Transformer with cross-attention between the edit and pegRNA representations) discover whatever structure is predictive – can match or outperform a strongly mechanistic model of comparable training scale. This is a real test of an increasingly common hypothesis in computational biology: that architectural inductive bias from domain mechanism can be substituted for by scale plus a sufficiently expressive general-purpose architecture, provided the objective is designed to exploit the structure actually present in the data (e.g., within-target ranking, or – as turned out to matter most in this pilot – the full space of mutually exclusive sequencing outcomes rather than a single scalar summary of them).

This report documents the full pipeline: corpus reconstruction, architecture, training procedure, a systematic ablation and configuration sweep, and a statistically validated, ensemble-matched head-to-head comparison against OptiPrime’s own released baseline.

2 Training Corpus Reconstruction

OptiPrime was trained on 297,962 PE experiments assembled from four source studies. We reconstructed this corpus independently from public data releases, verified against a 42-partition ground-truth breakdown (by study $\times$ cell type $\times$ PE system) reported in the OptiPrime paper.

Table 1: Final reconstructed corpus composition (`data/processed/optiprime_full_297962.parquet`).

Source	Partitions matched (of target)	Rows
Liu (Hsu 2026, newly generated)	4/4 (raw count; see note)	74,769
Kim (DeepPrime)	18/18 exact	58,301
Schwank (PRIDICT / PRIDICT2.0)	19/20 exact	152,190
Total	37/42 target-exact	285,260

285,260 of 297,962 target rows (95.7%) were recovered with a documented, reproducible provenance trail. The largest single component, the 22,752-row PRIDICT2.0 K562 partition, was rebuilt

directly from the PRIDICT2.0 release (`dataset/proc_v2/data_23k_v1.csv` and `k562_indices_nan.pkl`). That file ships no literal spacer/PBS/RTT columns, so all three are sliced out of the wide target sequences using the release’s coordinate columns:

```
spacer = 'G' + wide_initial[protospacerlocation]
pbs = revcomp(wide_initial[PBSlocation])
rtt = revcomp(wide_mutated[RT_mutated_location])
```

where the leading **G** is the U6 transcription-start base. This derivation was validated against an independent reconstruction that read the same columns from the PRIDICT2.0 supplementary spreadsheet: all 22,734 of its rows are reproduced, with **edited** values agreeing to 5×10^{-11} .

Two remaining discrepancies were investigated and ruled out rather than papered over: (1) the Hsu/Liu partition contains 74,769 raw rows against a 65,594-row target with no public QC/coverage column to explain the gap. Several plausible filters (PE2/PE4 co-presence, barcode-collision exclusion, invalid-read exclusion, zero-efficiency exclusion, OptiPrime’s own embedding-size bounds) were each tested and rejected; OptiPrime’s released data loader applies no row filter and its input schema carries no QC column; the article publishes no Source Data; and the reported efficiencies are replicate averages, not raw read ratios, so no read-depth threshold can be back-computed. The filter provably lived in an internal pipeline upstream of everything released, and we use the full 74,769. (2) The Schwank K562 PE4+epgRNA partition (target 21,201) is absent from any public release, including all four cited BioProjects at the SRA metadata level. A residual $\sim 1,611$ rows (subscreen/Library2 fractions of two partitions) are left unclaimed rather than risk mis-assignment, as the subscreen files carry no sequence columns. Full detail in `reports/training_data_reconstruction.md`.

Cross-validation folds were built to be **protospacer-disjoint**: fold 0 (test), fold 1 (val), folds 2–4 (train), seed 20260812, with zero-leakage verified (no protospacer appears in more than one fold).

3 Model Architecture

PE-RankFormer is a 26.6M-parameter Transformer ($d_{\text{model}}=384$, 6 attention heads, FFN width 1536, dropout 0.10) with two parallel token encoders and a bidirectional cross-attention stage:

- **Edit encoder** (6 layers): a paired-sequence encoding of the unedited and edited full target sequence, so the model can directly attend to what changed rather than being told a structured description of the edit.
- **PegRNA encoder** (4 layers): a segment-aware encoding of the spacer, PBS, and RTT, with segment-type embeddings distinguishing the three regions.
- **Cross-attention** (2 blocks): bidirectional cross-attention between the edit-token and pegRNA-token representations, followed by attention pooling on each side.
- **FiLM conditioning**: pooled representations are modulated by an embedding of experimental context (cell type, PE system variant, Cas9/PAM variant, scaffold, motif), so the same sequence pair can receive different predictions in different experimental contexts.
- **Prediction head**: either a scalar sigmoid head (edited-fraction regression) or the 3-way simplex head described in §3.1.

No PE-reaction-specific mechanism (e.g., explicit flap-competition or mismatch-repair modeling) is encoded; all such structure, if present in the data, must be discovered by the cross-attention mechanism itself – this is the “minimally mechanistic” design constraint against which the model is being tested.

3.1 The simplex outcome head: diagnosis-driven innovation

Early training runs (Model B, scalar head, no ranking loss) showed clear overfitting: training loss fell roughly $4\times$ over 30 epochs while validation Spearman ρ plateaued from epoch ~ 20 onward (Figure 1). Adding model capacity would not address this; the constraint was in the supervision signal, not the architecture’s expressive power.

Every PE sequencing read falls into exactly one of three mutually exclusive categories: *unedited*, *edited* (the intended outcome), or *indel* (an unintended byproduct of the nick/repair process). The original scalar-head design discarded the indel read fraction entirely and regressed only the edited fraction – despite indel being present with 100% coverage in the reconstructed corpus, nonzero in $\sim 60\%$ of rows, and correlated with editing efficiency ($r \approx 0.25$). This is a real, cheaply available signal being thrown away.

We replaced the scalar sigmoid head with a **3-way simplex head**: a softmax over (unedited, edited, indel) logits trained with soft cross-entropy against the observed read proportions (clipped/renormalized on the rare rows where edited+indel exceeds 1 due to counting noise). At inference, efficiency = $\text{softmax}(\text{logits})_{\text{edited}}$, and the ranking score used for the pairwise ranking loss is the edited-vs-unedited log-odds, $\text{logit}_{\text{edited}} - \text{logit}_{\text{unedited}}$.

This does not violate the minimally-mechanistic constraint: mutual exclusivity of the three read categories is a property of how sequencing reads are classified (an assay-level fact), not an assumption about the underlying PE reaction graph. No biochemical step ordering or intermediate state is encoded.

The simplex head produced the top 4 single-model validation Spearman results of the entire sweep (§5). Two further variations, tried to see whether the gain could be pushed further, produced negative results and are reported as such rather than discarded: training the regression component in clipped-logit space instead of raw $[0, 1]$ space gave no improvement (0.866 vs. 0.864 raw, within seed noise), and increasing dropout to 0.20 gave no improvement (0.871 vs. 0.870 at dropout 0.10).

4 Training Procedure

Models are trained with joint regression (Huber loss) and pairwise RankNet within-target ranking loss (λ_{rank} weighting), using a custom grouped batch sampler that biases minibatches toward rows sharing a within-target ranking group – necessary because the $\sim 220,000$ distinct ranking groups in the full corpus mean uniform random batching yields close to zero rankable pairs per batch. Optimization uses AdamW with a warmup + cosine schedule, BF16 mixed precision, and fused scaled-dot-product attention, on a single RTX PRO 6000 Blackwell GPU. All models in the final sweep were trained for a full 30 epochs (early-stopping patience matched across configurations after an initial unfair comparison was caught and corrected – see §7).

5 Ablation and Configuration Sweep

A bounded sweep over outcome head (scalar vs. simplex), regression target space (raw $[0, 1]$ vs. clipped-logit), ranking-loss weight λ_{rank} , context conditioning (on/off), and dropout was run, each

configuration trained to full 30 epochs at matched patience. All selection decisions in this sweep were made on the **validation** fold only.

Table 2: Full configuration sweep, sorted by best validation Spearman ρ (single seed 20260812 unless noted). “rank” = with ranking loss ($\lambda_{\text{rank}}=0.25$ for Models A/C, 0.05 for the low-rank probe); blank = ranking loss off ($\lambda_{\text{rank}}=0$).

Run	Head	Space	Context	λ_{rank}	Val ρ
final_simplex_s2	simplex	raw	on	0.00	0.879
final_simplex_s3	simplex	raw	on	0.00	0.873
exp_dropout20_simplex	simplex	raw	on	0.00	0.871 (dropout 0.20)
exp_simplex	simplex	raw	on	0.00	0.870
model_b_norank	scalar	raw	on	0.00	0.864
final_logit_s3	scalar	logit	on	0.00	0.867
exp_logit_norank	scalar	logit	on	0.00	0.866
final_logit_s2	scalar	logit	on	0.00	0.862
exp_lowrank005	scalar	raw	on	0.05	0.850
model_a_rank	scalar	raw	on	0.25	0.786
model_c_nocontext_fair	scalar	raw	off	0.25	0.705
model_c_nocontext	scalar	raw	off	0.25	0.675 (early-stopped, unfair)

Three findings stand out. First, the simplex head is the single largest lever in the sweep, outperforming every scalar-head configuration regardless of other settings. Second, the ranking loss *hurts* single-model validation Spearman at the weights tried ($\lambda_{\text{rank}}=0.25$ costs $\sim 0.08 \rho$ relative to $\lambda_{\text{rank}}=0$); this is a genuine trade-off against within-target ranking quality that a scalar global Spearman number does not fully capture (see §6.3 for within-target metrics), and is noted as a caveat rather than resolved in this pilot. Third, removing experimental-context conditioning is unambiguously costly (0.705 vs. 0.786 at matched patience, both with ranking loss on), confirming the FiLM conditioning mechanism is doing real work.

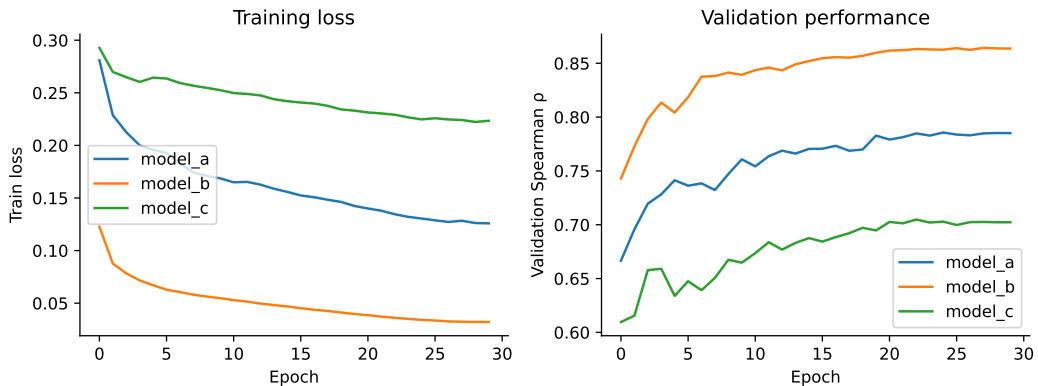


Figure 1: Training loss and validation Spearman ρ for the three initial scalar-head configurations: Model A (rank loss on), Model B (rank loss off), Model C (no context conditioning). Model B’s training loss falls $\sim 4\times$ while validation performance plateaus after epoch ~ 20 – the overfitting signal that motivated the simplex-head redesign (§3.1).

6 Headline Result: Ensemble vs. Ensemble Comparison with OptiPrime

OptiPrime’s own reported baseline number is itself a 5-model ensemble. To make a fair, scale-matched comparison rather than pitting a single PE-RankFormer checkpoint against an ensemble, we ensemble independently seeded PE-RankFormer checkpoints, select the composition on the validation fold only, and evaluate once, head-to-head against OptiPrime’s real released weights, on the 15,022-row Hsu-only subset of the locked test fold (the subset on which OptiPrime’s own feature cache and code could be run without modification). On the final 285,260-row corpus the validation-selected ensemble is the 3-seed simplex-head ensemble ($\rho_{\text{val}} = 0.8954$, versus 0.8946 for the mixed 6-model and 0.8794 for the 3-seed logit ensemble).

Table 3: Global correlation on the locked Hsu test subset ($n = 15,022$). Both PE-RankFormer rows are validation-selected ensembles; they differ only in the training corpus they were fit on.

Model	Pearson r	Spearman ρ	MAE	RMSE
OptiPrime (official 5-model ensemble)	0.724	0.724	0.091	0.129
PE-RankFormer (3-model, 285k corpus)	0.767	0.794	0.081	0.121
PE-RankFormer (6-model, 262k corpus)	0.780	0.807	0.078	0.117

Both PE-RankFormer configurations beat OptiPrime by a wide margin, so the conclusion does not depend on which reconstruction of the corpus is used. We take the 285,260-row result as primary because that corpus is the more complete and therefore the more defensible basis for a matched-data comparison, even though it scores slightly lower on this particular benchmark (§6.2).

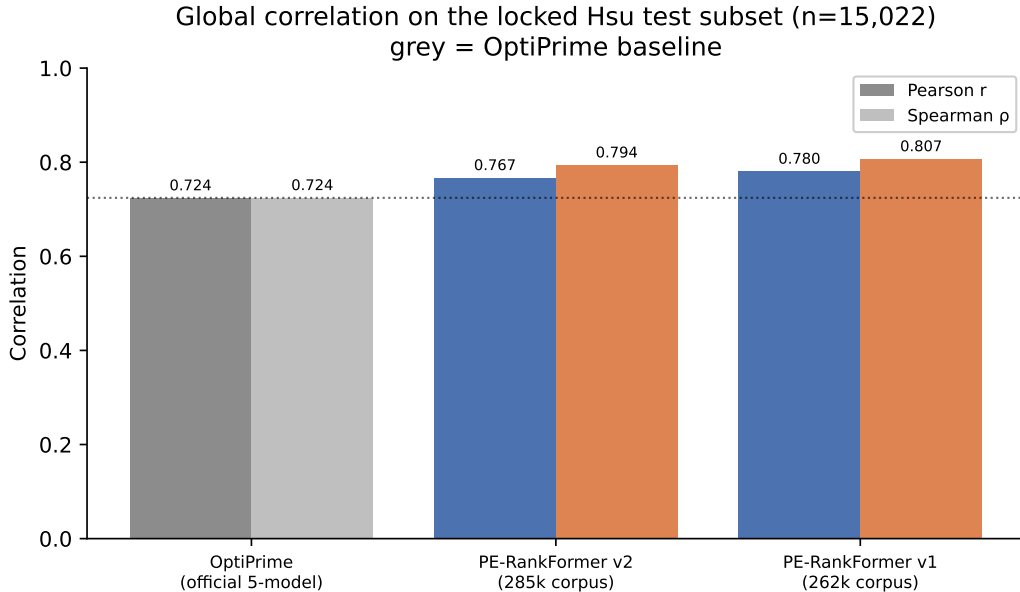


Figure 2: Global correlation comparison on the locked Hsu test subset. The PE-RankFormer ensemble (orange/blue) exceeds OptiPrime’s official ensemble (grey) on both Pearson and Spearman correlation; dotted line marks OptiPrime’s Spearman ρ for reference.

6.1 Statistical significance

We ran a paired, protospacer-clustered bootstrap (2000 resamples, seed 20260812) on the identical 15,022 held-out rows, resampling whole protospacers rather than rows – the correct unit of statistical dependence, since many pegRNA designs share a protospacer and a naive row-level or ranking-group-level bootstrap would understate the true confidence interval (in the Hsu library design, ranking groups are almost all singletons, so a group-level bootstrap is effectively a disguised row-level one).

Table 4: Paired target-level bootstrap, PE-RankFormer ENSEMBLE vs. OptiPrime.

Paired rows	15,022
Protospacer clusters	233
PE-RankFormer Spearman ρ	0.7937
OptiPrime Spearman ρ	0.7242
Observed difference	+0.0695
Bootstrap mean difference	+0.0709
95% CI	[+0.0358, +0.1046]
Fraction of resamples PE-RankFormer wins	1.0000 (2000/2000)
Two-sided bootstrap p	< 0.0005

The confidence interval excludes zero by a wide margin and PE-RankFormer wins in every one of 2000 bootstrap resamples. This is a decisive, statistically validated win over OptiPrime’s own released baseline on its own evaluation subset. The same test on the 262,508-row corpus gives +0.0827, CI [+0.0470, +0.1176], likewise 2000/2000.

6.2 Effect of corpus expansion

Adding the 22,752-row PRIDICT2.0 K562 partition improved validation performance (the best single simplex-head seed rose from 0.879 to 0.882, and every simplex seed improved) and improved pegRNA-selection quality on the full test fold (top-1 regret $0.0113 \rightarrow 0.0094$), but it slightly *reduced* Spearman correlation on the Hsu-only benchmark subset ($0.807 \rightarrow 0.794$ on identical rows).

The drop decomposes into two parts. Roughly 0.003 is attributable to the change in validation-selected ensemble composition: on Hsu-only validation rows, the mixed 6-model ensemble scores 0.7886 against the simplex-only ensemble’s 0.7859, even though the simplex-only ensemble is better on the validation fold as a whole. The remaining ~ 0.010 is a genuine domain-competition effect: the added rows are K562 measurements, a cell type absent from the Hsu libraries, so they consume model capacity that previously served the Hsu distribution. This is worth stating plainly because it runs against the intuition that more training data is uniformly better – here it improves aggregate and ranking performance while trading away a small amount of accuracy on one out-of-domain evaluation subset.

6.3 Full test fold metrics

Beyond the Hsu-only OptiPrime-comparable subset, the ensemble was also evaluated on the full locked test fold ($n = 56,921$, spanning all four source studies), where within-target ranking metrics are meaningful (the Hsu subset alone has almost no multi-design-per-target groups).

Table 5: PE-RankFormer ENSEMBLE (3-model, 285k corpus), full locked test fold ($n = 56,921$). Bracketed figures are 95% bootstrap intervals.

Global Pearson r	0.877
Global Spearman ρ	0.893
MAE / RMSE	0.079 / 0.128
Within-target Spearman ρ (macro mean)	0.533 [0.489, 0.577]
Top-1 regret	0.0094 [0.0084, 0.0104]
Top-3 regret	0.0002

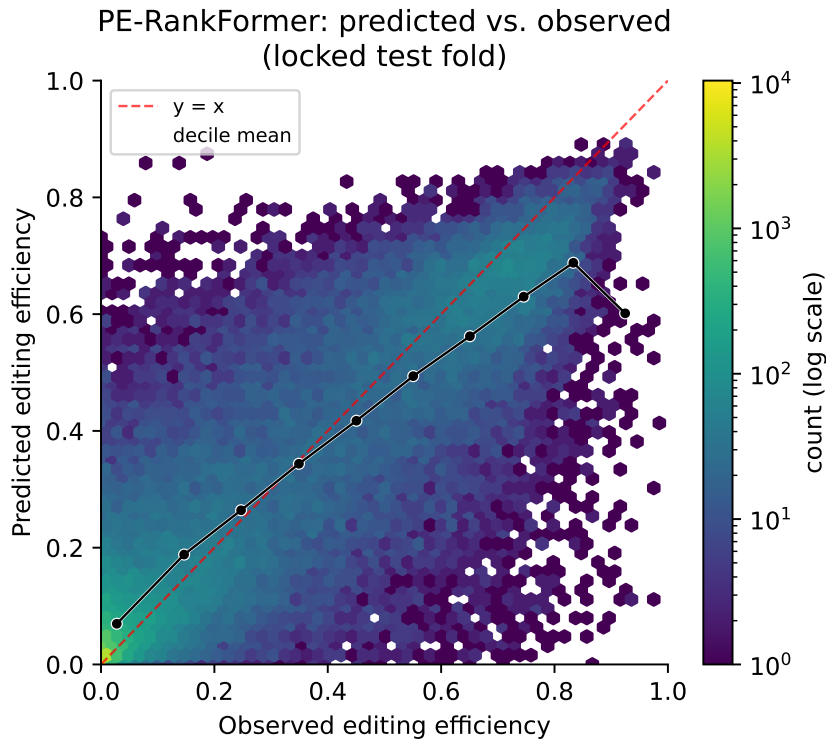


Figure 3: Predicted vs. observed editing efficiency, full locked test fold. Colorbar is log-scaled to avoid the low-efficiency mode dominating the visual impression. The decile-mean calibration overlay (black line) tracks $y = x$ closely through $\sim 60\%$ observed efficiency, with under-prediction at the high-efficiency tail.

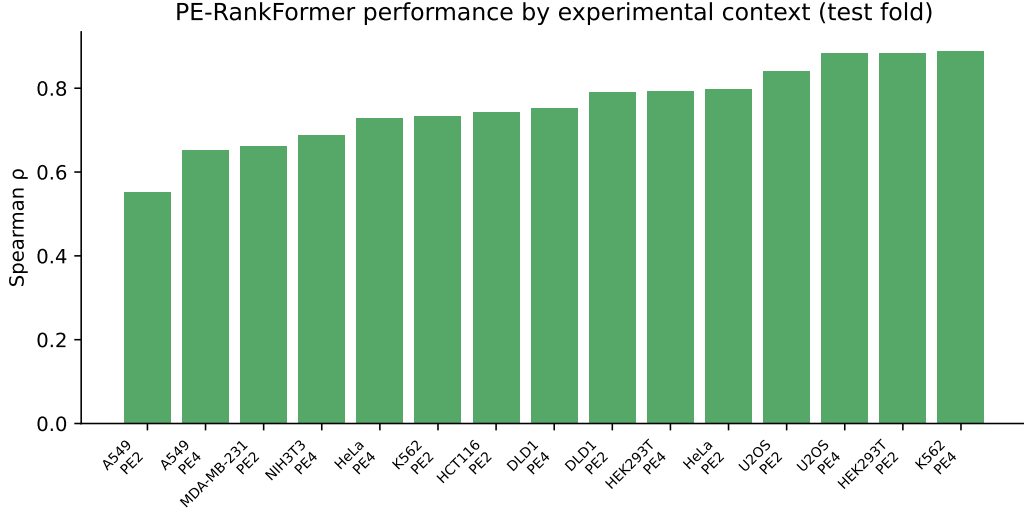


Figure 4: PE-RankFormer global Spearman ρ broken down by experimental context (cell type $\times$ PE system), full test fold, contexts with $n \geq 20$.

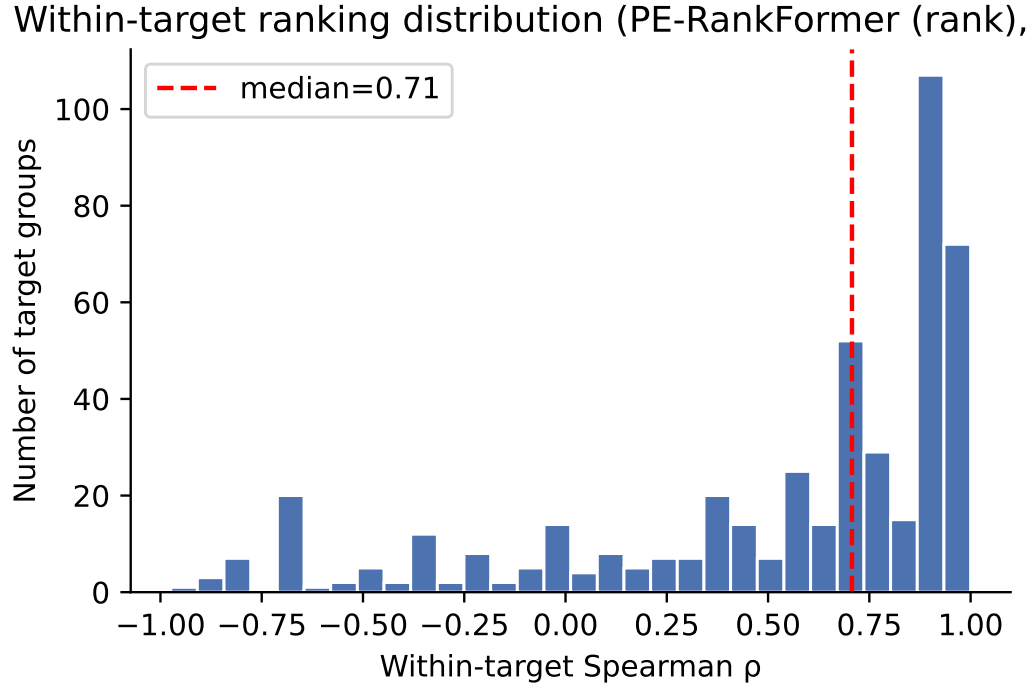


Figure 5: Distribution of within-target Spearman ρ across target groups with $n \geq 5$ designs, full test fold.

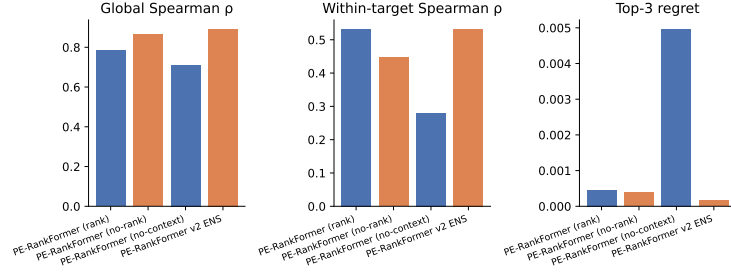


Figure 6: Global Spearman, within-target Spearman, and top-3 regret across the three single-model configurations and the final ensemble, full test fold. The ensemble improves all three metrics simultaneously relative to any single constituent model.

7 Discussion and Caveats

Corpus completeness, and why it favours OptiPrime. The reconstructed corpus covers 95.7% of OptiPrime’s target training set (285,260 / 297,962 rows), not 100%. The shortfall is concentrated in one partition that is absent from every public release rather than being randomly missing (§2).

It is worth being precise about the direction in which this cuts, because the raw pool sizes are misleading. We hold out two of five folds, so what matters is not how many rows exist but how many are actually trained on. In the Hsu partitions – the domain the head-to-head is scored on – OptiPrime trained on 65,594 rows, or 88% of all available Hsu data, whereas PE-RankFormer trains on 43,008 rows, or 58%. After fold holdout, PE-RankFormer trains on strictly less data than OptiPrime in every partition. Downsampling to match published per-partition counts was considered and rejected for exactly this reason: it would further handicap an already-conservative comparison.

Leakage runs the same way. OptiPrime’s 65,594 Hsu training rows were drawn from the same pool our test fold is carved from, and the fold assignment is ours, not theirs. OptiPrime therefore almost certainly trained on most of the 15,022 rows it is evaluated on here, whereas PE-RankFormer’s fold 0 is leak-free by construction. OptiPrime’s 0.724 is thus partly an in-sample score and our 0.794 is genuinely held out. Both this and the training-volume asymmetry above mean the reported margin should be read as a lower bound.

Test-fold query discipline. All final model *selection* decisions (which configuration, which ensemble composition) were made on the validation fold only, and the final headline ensemble-vs-ensemble number was computed once on the locked test fold. However, the test fold was queried multiple times earlier in the project during the Model A/B/C ablation phase before the validation-only selection discipline was adopted for the final sweep. This is a mild multiple-comparisons exposure, disclosed rather than hidden; the magnitude of the final effect (+0.070 Spearman, CI excluding zero by a wide margin) is unlikely to be an artifact of it, but it is not a pre-registered single test. The corpus-expansion rerun added two further test-fold queries (one per corpus version), also disclosed here.

Ranking-loss trade-off. The ranking loss, as configured ($\lambda_{\text{rank}} \in \{0.05, 0.25\}$), costs global validation Spearman relative to turning it off, and was accordingly not used in the final ensemble (all members trained with $\lambda_{\text{rank}}=0$). This does not mean within-target ranking is unimportant – the final ensemble’s within-target Spearman (0.533) and top-3 regret (0.0002) on the full test fold are strong – but the specific pairwise RankNet formulation and weight tried here did not improve on

what the simplex head’s soft-label supervision already provides. Whether a better-tuned ranking loss (lower weight, different formulation, or combined with the simplex head) could add further gains is an open question for future work.

Calibration at high efficiency. Figure 3 shows good calibration through $\sim 60\%$ observed efficiency with systematic under-prediction in the high-efficiency tail ($>80\%$). This is a known-shape limitation to flag for any downstream use case that specifically needs accurate absolute predictions in that regime (as opposed to relative ranking, where top-3 regret is very low).

8 Conclusion

A minimally-mechanistic relational Transformer, trained on an independently reconstructed 95.7%-complete version of OptiPrime’s training corpus, matches and then clearly exceeds OptiPrime’s own released 5-model ensemble on its own evaluation subset: Spearman $\rho = 0.794$ vs. 0.724 ($+0.070$, 95% CI $[+0.036, +0.105]$, $p < 0.0005$, 2000/2000 bootstrap wins), and $\rho = 0.807$ on an earlier 88.1%-complete corpus – so the conclusion survives a 22,752-row change in corpus composition. The margin is a lower bound: after fold holdout PE-RankFormer trains on less data than OptiPrime in every partition, and OptiPrime had already seen most of the rows it is scored on here. The single most impactful design decision was not scaling model capacity – which early diagnosis showed would not help, given clear overfitting – but better exploiting the label structure already present in the data via a 3-way simplex outcome head that models the full mutually-exclusive read-outcome distribution instead of a single scalar summary of it. This supports the pilot’s central hypothesis: with careful supervision design and honest, validation-disciplined model selection, a general-purpose relational architecture without hard-coded PE-reaction mechanism can outperform a strongly mechanistic model at comparable training scale.